

Physics lab 4

August 15, 2026

Q1: Interference

Method: We simulated the time-dependent Schrödinger equation for a Gaussian wave packet reflecting off an infinite potential wall. The simulation was run at two different mesh resolutions ($N = 2^{13}$ and $N = 2^{14}$) to observe the resulting probability density as the incident and reflected waves overlap.

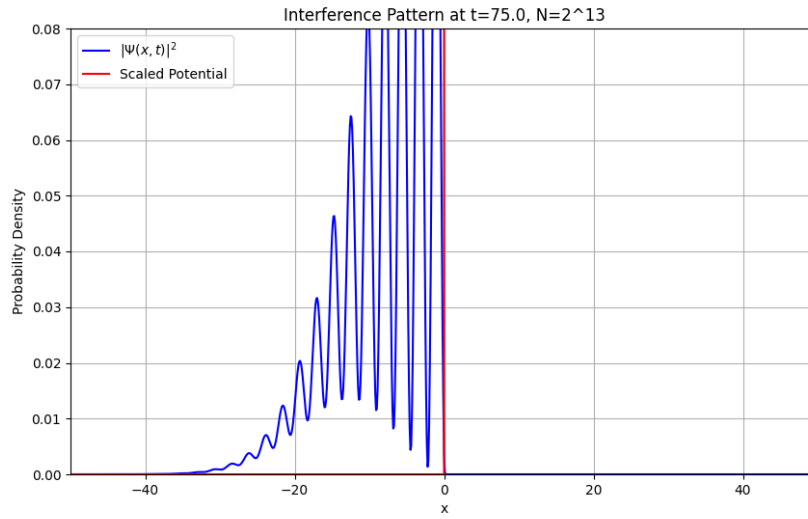


Figure 1: Interference Pattern ($N = 2^{13}$)

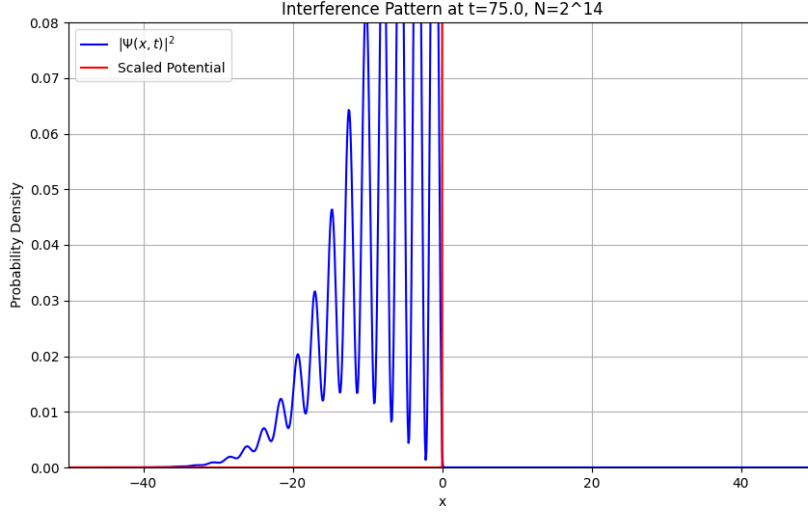


Figure 2: Interference Pattern ($N = 2^{14}$)

Theory: According to quantum mechanical theory, we expect the incident and reflected wave packets to superimpose, creating an interference pattern with spatial oscillations in the probability density $|\Psi|^2$. Since this is a physical wave phenomenon, it should be independent of the numerical grid resolution.

Results: The plots show a distinct interference pattern during the reflection. The pattern remains identical across both mesh resolutions ($N = 2^{13}$ and $N = 2^{14}$), which confirms that the interference is a real physical phenomenon and not a numerical artifact of the simulation grid.

Q2: Gamow Tunneling

Method: We simulated the splitting of a wave packet colliding with the Gamow Coulomb barrier $V = \begin{cases} -V_0 & \text{for } x < R \\ C/x & \text{for } x > R \end{cases}$ where $C = R = 1$ and $V_0 = 0$ with $E = 0.9$. We then calculated the transmission probability T for several initial energies E and plotted $\ln T$ against $1/\sqrt{E}$.

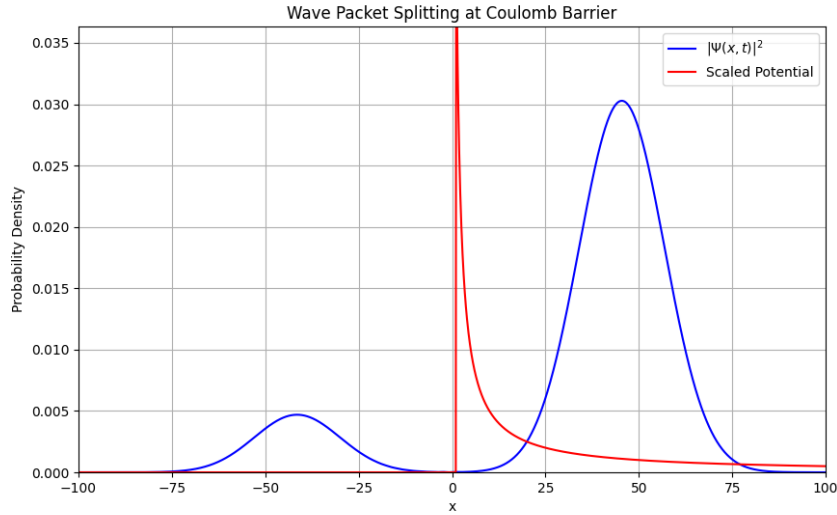


Figure 3: Wave Packet Splitting at Coulomb Barrier, $E = 0.9$

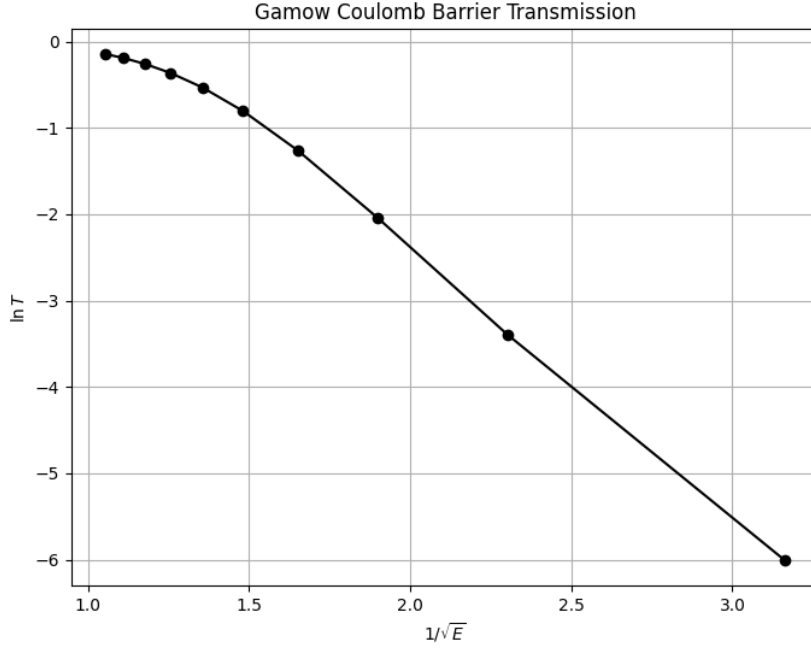


Figure 4: Transmission Probability T vs kinetic energy $E \in [0.1, 0.9]$

Theory: According to Gamow’s theory of alpha decay, quantum tunneling through a Coulomb barrier predicts a transmission probability that depends exponentially on energy. Specifically, the theory states that $T \propto \exp(-k/\sqrt{E})$, which implies that a plot of $\ln T$ versus $1/\sqrt{E}$ should yield a straight line with a negative slope. Consequently, from the approximation $T = \frac{Y_T}{Y_T + Y_R}$, at high E we expect T to also be high, which requires Y_T to be much larger than Y_R (i.e. the transmitted wave packet is much larger than the reflected one).

Results: In fig 3, we observe that the wave packet splits into a reflected and transmitted component when encountering the Coulomb barrier, confirming the tunneling effect predicted by quantum mechanics. We also observe that the transmitted wave packet is significantly smaller than the reflected one, consistent with the high transmission probability at high energies.

In fig 4, the plot of $\ln T$ against $1/\sqrt{E}$ shows a clear linear relationship at lower energies. This linear portion perfectly agrees with Gamow’s theoretical exponential relationship for alpha decay tunneling.

Q3: Fowler-Nordheim Tunneling

Method: We initialized a Gaussian wave packet with width $a = 10.0$, centered at $x_0 = -100.0$, and an initial energy $E = 0.5$. The simulation utilized a spatial grid of $N = 2^{14}$ points with a step size $dx = 0.1$ and a time step $dt = 0.01$. We set the reduced Planck constant $\hbar = 1.0$ and the particle mass $m = 1.0$.

First, we simulated the wave packet encountering a flat potential step $V(x) = V_0 = 1.0$ for $x > 0$ to observe total reflection, since $E < V_0$. Then, we simulated field emission tunneling through a slanted potential barrier $V(x) = V_0(1 - x/L)$ for $0 < x < L$ with $L = 4.0$.

We then calculated the transmission T past $x = 4.0$ for varying barrier heights $V_0 \in [0.6, 1.0]$ and plotted $\ln T$ against $V_0^{3/2}$.

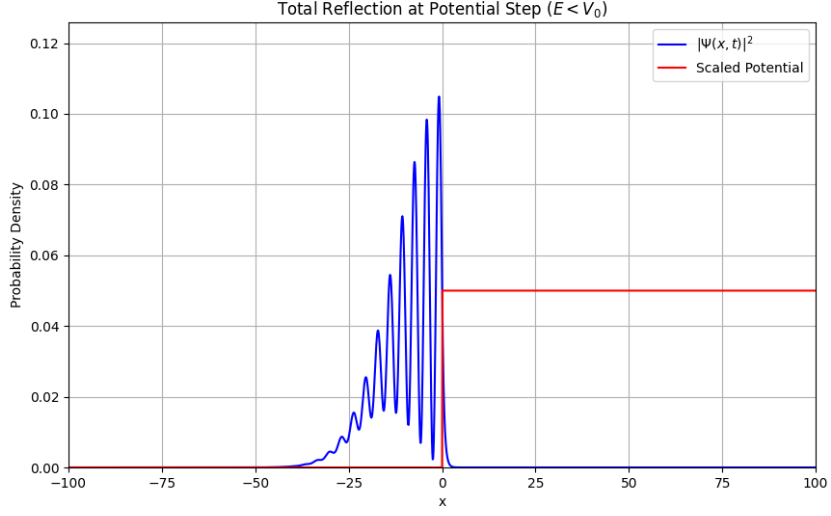


Figure 5: Complete Reflection at Flat Potential Step, $V_0 = 1.0$, $L = 4.0$, $E = 0.5$

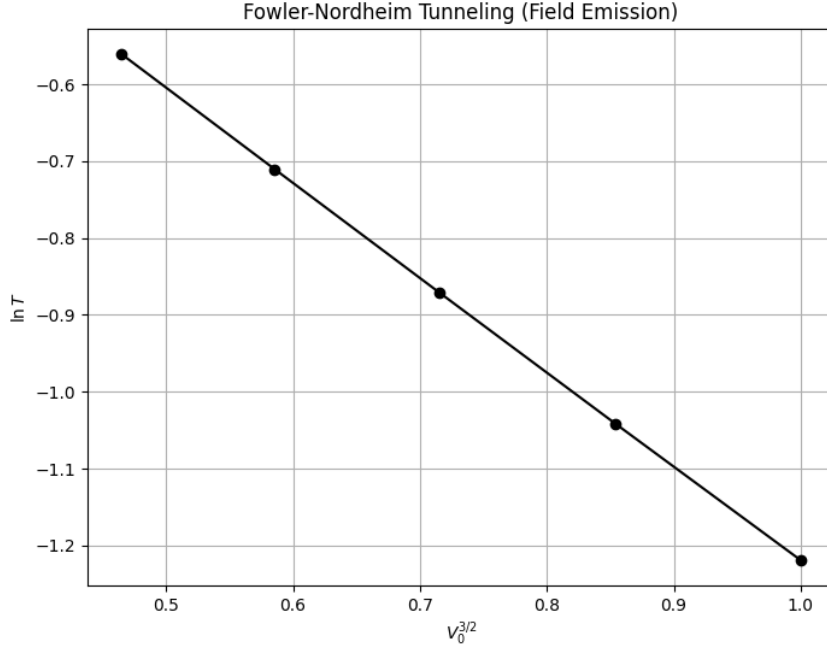


Figure 6: Fowler-Nordheim Tunneling, $\ln T$ vs $V_0^{3/2}$ for $V_0 \in [0.6, 1.0]$

Theory: According to the Fowler-Nordheim theory of field emission, the transmission probability through a triangular potential barrier is given by $T \propto \exp(-c \cdot V_0^{3/2}/F)$. Therefore, we expect a plot of $\ln T$ versus $V_0^{3/2}$ to yield a straight line.

Results: Figure 5 demonstrates complete reflection at the flat potential step, as expected classically when $E < V_0$. Figure 6, the plot of $\ln T$ against $V_0^{3/2}$ for the slanted potential, shows a definitive linear relationship, which firmly confirms that our numerical simulation matches the Fowler-Nordheim tunneling theory.